

QN 1 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 1.defaultsorts.cpp -o 1.defaultsorts
age/fifthSem/DAA/"1.defaultsorts

Sorting using Bubble Sort
Before sorting:
100    22    12    45    809    130    170    10    200
After sorting:
10     12    22    45    100    130    170    200    809
No. of Steps required : 612

Sorting using Selection Sort
Before sorting:
10     12    22    45    100    130    170    200    809
After sorting:
10     12    22    45    100    130    170    200    809
No. of Steps required : 288

Sorting using Insertion Sort
Before sorting:
10     12    22    45    100    130    170    200    809
After sorting:
10     12    22    45    100    130    170    200    809
No. of Steps required : 56
```

QN 2 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 2.mergesort
fifthSem/DAA/"2.mergesort

Before sorting:
100    22    12    45    809    130    170    10    200
After sorting:
10     12    22    45    100    130    170    200    809
No. of Steps required : 367
```

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
```

QN 3 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*  
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 3.quick  
fthSem/DAA/"3.quicksort
```

Sorting using Quick Sort , case = Average Case

Before sorting:

100	22	12	45	809	130	170	10	200
-----	----	----	----	-----	-----	-----	----	-----

After sorting:

10	12	22	45	100	130	170	200	809
----	----	----	----	-----	-----	-----	-----	-----

No. of Steps required : 145

Sorting using Quick Sort , case = Worst Case

Before sorting:

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

After sorting:

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

No. of Steps required : 220

QN 4 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*  
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 4.r_qui  
e/fifthSem/DAA/"4.r_quicksort
```

Sorting using Randomized Quick Sort , case = Average Case

Before sorting:

100	22	12	45	809	130	170	10	200
-----	----	----	----	-----	-----	-----	----	-----

After sorting:

10	12	22	45	100	130	170	200	809
----	----	----	----	-----	-----	-----	-----	-----

No. of Steps required : 169

Sorting using Randomized Quick Sort , case = 'Worst' Case

Before sorting:

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

After sorting:

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

No. of Steps required : 167

QN 5 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
> cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 5.knap:
age/fifthSem/DAA/"5.knapsack_0_1
Enter the capacity and number of items: 7 4
Enter weight and value of item [1]: 1 3
Enter weight and value of item [2]: 2 4
Enter weight and value of item [3]: 5 2
Enter weight and value of item [4]: 5 7

Knapsack DP Table:
0      0      0      0      0      0      0      0
0      3      3      3      3      3      3      3
0      3      4      7      7      7      7      7
0      3      4      7      7      7      7      7
0      3      4      7      7      7      10     11
```

QN 6 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main
> cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 6.mat:
t/Development/Programming/Collage/fifthSem/DAA/"6.mat:
Enter number of matrices: 4
Enter 5 dimensions: 3 4 5 2 6

Optimal Parenthesization: ((A1(A2A3))A4)

Minimum number of multiplications: 100
Number of steps required: 220
```

QN 7 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
> cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 7.string_editing
Collage/fifthSem/DAA/"7.string_editing
Enter first string: Bsc.CsiT
Enter second string: Bsc.It
Maximum no. of edit required : 4
No. of steps Required : 763
```

QN 8 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/"
  lopment/Programming/Collage/fifthSem/DAA/"8.Floyd_Warshall
The following matrix shows the shortest distances between
0 5 8 9
8 0 3 4
5 10 0 1
8 13 3 0
Required No of steps : 378
```

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
○ >
```

QN 9 Output:

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 9
  pment/Programming/Collage/fifthSem/DAA/"9.Dijkstras_single_source
Vertex    Distance from Source
0
1
2
3
4
5
6
7
8
14
No. of steps required :191
```

QN 10 Output

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/"
  ing/Collage/fifthSem/DAA/"10.N_Queen_Problem
..Q.
Q...
...Q
.Q..
No. of required steps: 381
```

QN 11 Output

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 11.Kruskal's_Algorithm.cpp
Edges of MST:
6 - 9 (Weight=1)
1 - 7 (Weight=1)
7 - 11 (Weight=2)
0 - 10 (Weight=2)
4 - 9 (Weight=2)
2 - 6 (Weight=3)
2 - 8 (Weight=3)
3 - 5 (Weight=3)
3 - 6 (Weight=3)
7 - 8 (Weight=3)
0 - 11 (Weight=3)

Total Weight of MST: 26
```

QN 12 Output

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/" && g++ 12.Prims_min.cpp
Edge    Weight
0 - 1    2
4 - 2    9
2 - 3    6
1 - 4    7
3 - 5    6
9 - 6    5
8 - 7    5
0 - 8    3
5 - 9    4
```

QN 13 Output

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*  
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/"  
e/fifthSem/DAA/"13.Subset_Sum  
8 9 14 22  
8 14 15 16  
15 16 22  
Nodes generated: 68
```

QN 14 Output

```
/mnt/Development/Programming/Collage/fifthSem/DAA main*  
• > cd "/mnt/Development/Programming/Collage/fifthSem/DAA/"  
t/Development/Programming/Collage/fifthSem/DAA/"14.Job!  
Maximum Profit Job Sequence:  
c a e  
Total Profit: 142
```